MODULE VI: TEMPERATURE RESPONSIVE VVI-R CARDIAC PACEMAKER
FINAL REPORT

ELLA FORTERNBERY & ROSEMINA AYIEKO
GROUP M10

INTRODUCTION

Background

The heart relies on a coordinated electrical conduction system to maintain effective and rhythmic contractions. In a healthy heart, impulses originate in the sinoatrial (SA) node and propagate through the atria to the atrioventricular (AV) node, then through the His–Purkinje system to stimulate ventricular contraction. This sequence ensures proper timing of blood flow and allows heart rate to adjust to physiological demand [1].

Disruptions in this system can lead to arrhythmias that impair cardiac function. Bradycardia, caused by failure of impulse generation or conduction (e.g., SA node dysfunction or AV block), results in abnormally slow heart rates and reduced cardiac output. Conversely, abnormal conduction pathways can produce tachycardia or irregular rhythms that diminish the efficiency of coordinated contraction [3].

When the heart cannot reliably generate or conduct electrical signals, a cardiac pacemaker is required to restore rhythm. Pacemakers deliver electrical stimuli to ensure timely depolarization when intrinsic activity is absent or insufficient, thereby maintaining adequate perfusion [1].

Although conventional pacemakers effectively prevent dangerously low heart rates, they typically pace at a fixed rate and do not adapt to changes in physiological demand. In a healthy system, heart rate rises with metabolic activity such as exercise or fever. Rate-responsive pacemakers, denoted by the trailing "R" in standard NBG nomenclature (e.g., VVI-R) , address this limitation by using physiological sensors such as motion, minute ventilation, or temperature to modulate pacing rate in real time [2], [4].

Approach

The device developed in this work is a **VVI-R pacemaker**: it senses ventricular activity, paces only when the intrinsic rhythm falls below a programmed lower rate, and adapts its pacing rate to physiological state. The sensing chain consists of an instrumentation preamplifier, bandpass filter, gain stage, and comparator feeding an Arduino microcontroller. The microcontroller performs R–R interval detection, enforces lower-rate-interval (LRI) logic, and triggers a transistor-based stimulation circuit when an intrinsic beat is not detected within the LRI.

The rate-responsive functionality is implemented through a thermistor-based voltage divider that allows the microcontroller to estimate body temperature and modulate pacing rate accordingly. For temperatures above the febrile threshold of 38 °C, the device increases the pacing rate by approximately **12.5 BPM per degree of elevation**, capped at 180 BPM, mimicking the physiological tachycardia associated with fever [2]. An OLED display reports temperature, pacing heart rate, and an indication of fever state in real time.

PACEMAKER DESIGN

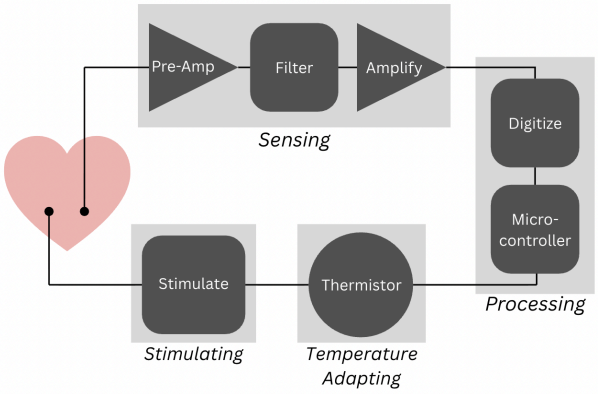


Figure 1. Block Diagram showing functionality of full VVI Pacemaker integrated circuit.

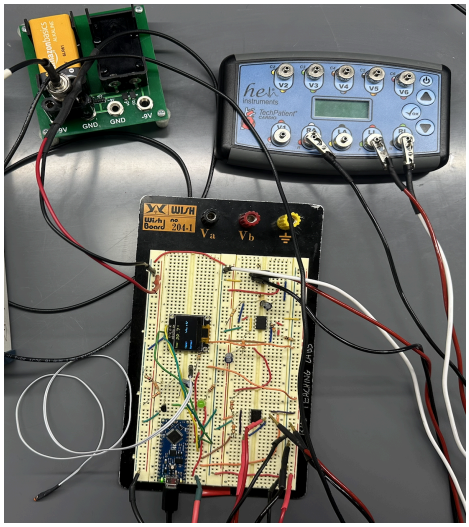


Figure 2. Photograph of fully connected breadboard circuit.

The pacemaker is implemented as four functional blocks: an analog sensing front end, a digital acquisition and processing subsystem, a stimulation output stage, and a temperature based rate-adaptation circuit. A block diagram of the full system is shown in **Figure 1**, and a photograph of the assembled breadboard prototype is shown in **Figure 2.** There is a fully detailed schematic in **Appendix A.1**.

Single-Supply Powering

All op-amps in the circuit are powered by 5V Single-Supply Voltage, delivered by a KEITHLEY Triple Channel DC Power Supply. To keep both negative and positive voltages from our acquired electrogram signal, the signal is offset by a constant 2.5 V. This reference voltage is created using a voltage divider and buffer from our power line, and is shown in **Appendix A.2**.

Sensing Front End

The sensing chain conditions the millivolt-scale electrogram for digital comparison. The differential ECG input is first amplified by an instrumentation amplifier **AD623 with RG = 0 Ωs,** and therefore no gain. This op-amp is chosen for its high common-mode rejection ratio. The amplified signal then passes through a passive bandpass filter formed by a high-pass section (cutoff fL ≈ 2 Hz) and a low-pass section (cutoff fH ≈ 60 Hz), defining a passband matched to the dominant spectral content of the QRS complex. The non-inverting side of an MCP6002 op-amp with **gain =1001** (since Rin= 1000Ωs and Rf =1MΩ), brings the QRS peak into a usable digital range. The total measured end-to-end gain was 589.3 V/V (see **Table 1**). The same MCP6002 is then used as a comparator to detect the QRS complex in the ECG. The threshold is set using a Wave Station 10 MHz Waveform Generator in order to be adjustable. **Appendix A.3** shows the zoomed schematic with these components.

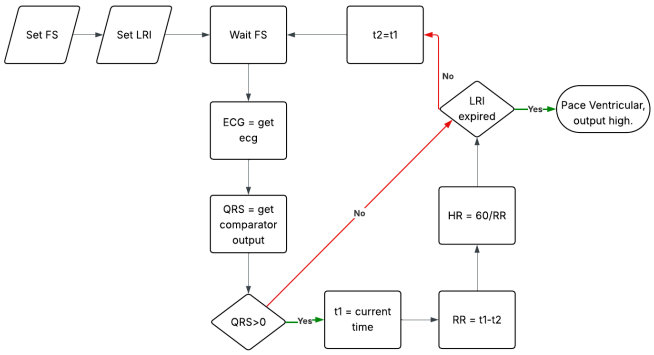


Figure 3. Flowchart showing how the microcontroller determines when to stimulate the ventricle.

Acquisition and Processing

The digital subsystem is implemented on an Arduino Nano Every. The comparator output is sampled at a user-defined sampling rate; for the characterization reported here, a sampling period of 1 ms (1000 samples/s) was used. Rising edges of the comparator pulse are time-stamped to compute R–R intervals in real time. The analog voltages are automatically converted by the Arduino Nano Every's analog-to-digital-converter (ADC). Therefore, we had to reconvert to a 0-5 V scale before using the values. The microcontroller maintains a running estimate of intrinsic heart rate and compares the most recent R–R interval to the programmed lower-rate interval (LRI). When the LRI is exceeded i.e., when no intrinsic beat has occurred within the maximum allowed interval , the arduino issues a digital pacing pulse of duration t = 2·chronaxie to the stimulation block. The arduino also stores these time values to calculate the instantaneous heart rate. This process is shown in **Figure 3**. This R–R determination accuracy was validated against a benchtop ECG simulator across 60, 90, 120, and 220 BPM with both 0% and 10% R–R variability over 10-second collection windows; results are summarized in **Table 2**.

Stimulation Output

Pacing pulses are delivered through a 2n 7000 Enhancement Mode n-type MOSFET-driven capacitor-discharge stimulation circuit. A storage capacitor C = 10 µF is charged through Rc = 82 kΩ to the supply voltage V_dd ~9.56V; when the Arduino issues a digital pacing trigger to the MOSFET gate, the capacitor discharges across the load resistance RL, delivering a stimulus pulse of magnitude approximately 9.56 V_dd to the load. For benchtop characterization, RL = 82.2 kΩ was used as a resistive proxy in place of the frog myocardium load (~1 kΩ) (which was actually applied during the final demo) , since end-to-end frog stimulation was not performed in this experiment. Pulse duration is software-set to t_pulse = 2·chronaxie = 0.4 ms, and the strength–duration parameters that determine this are derived by manually characterizing working stimuli on a frog's myocardium. We found the rheobase to be b = 1.5 µA and chronaxie to be c = 0.2 ms, see **Figure 9**. Stimulating at I_{stim} > 2b for d_{stim} > 2c provides a safety factor over the threshold curve, ensuring reliable myocardial capture without excess charge delivery. Since the resistance of the myocardium is only ~1kΩ, we kept the 82kΩ in series to limit the current flowing to 100µA. This schematic is shown in **Appendix A.4.**

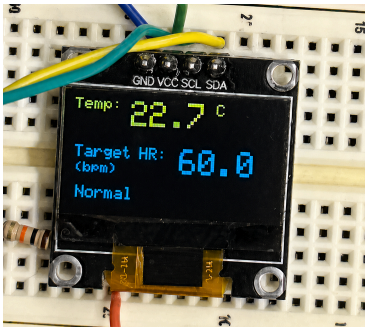


Figure 4. Photograph of working OLED display, when thermistor is exposed to room temperature.

Rate-Responsive Block

Body temperature is measured by a negative-temperature-coefficient (NTCLE100E3) thermistor configured as the upper leg of a voltage divider, with a fixed 10 kΩ reference resistor as the lower leg and V_in = 5 V supply. The divider output is read by an Arduino analog input and converted to absolute temperature by an extended four-coefficient Steinhart–Hart formulation [6]:

$$T = \frac{1}{A + B\ln\left(\frac{R}{R_{\text{ref}}}\right) + C\ln^2\left(\frac{R}{R_{\text{ref}}}\right) + D\ln^3\left(\frac{R}{R_{\text{ref}}}\right)}$$

The microprocessor compares the calculated temperature T against a programmed baseline temperature of 36 °C and an absolute febrile threshold of 38 °C; for T > 38 °C the pacing rate is set according to the following formula [2]:

$$\text{bpm} = \text{bpm}_i + 12.5\,\Delta T$$

We tested this functionality with an initial pacing rate of 60bpm, and the lower-rate interval is updated as LRI = 60000/bpm ms. A hard ceiling of **180 BPM** is enforced in software as a safety limit: regardless of how high the thermistor temperature reads, the pacing rate cannot exceed this value, preventing overstimulation and limiting the device's worst-case behavior should the thermistor or its calibration fail. A 0.96-inch I²C OLED display (SSD1306-based, address 0x3C) reports the calculated temperature, the current target BPM, and a Normal/Fever status label, providing real-time visual confirmation of system behavior during testing. This is shown in **Figure 4**. Both of these components have detailed schematics shown in **Appendix A.5**.

Block Characterization and Integration Testing

Each block was characterized in isolation before full-system integration. The sensing front end was characterized by sweeping a function-generator sinusoid and measuring the magnitude transfer function for the high-pass, low-pass, and combined bandpass stages. End-to-end gain was verified empirically with the ECG simulator. The comparator was bench-tested by feeding the simulator-derived electrogram into the threshold input and confirming that digital pulses tracked QRS events. Acquisition accuracy was validated as described above. The stimulation block was characterized by measuring delivered pulse amplitude and duration on the oscilloscope. The rate-responsive block was validated by sweeping temperature past the 38 °C threshold and recording the BPM response. Full-system integration was verified using the live frog electrogram as input.

TESTING & RESULTS

The pacemaker was characterized at the block level and as an integrated system. All circuit-level data were acquired with a benchtop ECG simulator providing controlled inputs; the full-system test used the frog electrogram as a physiological input.

PM Characterization

Electrogram Sensing: Transfer Functions, Gain, Frog ECG and Comparator

Electrogram Sensing

The high-pass and low-pass filter transfer functions are shown in **Figures 5 and 6**, respectively, and their cascaded bandpass response is shown in **Figure 7**. The combined response demonstrates selective amplification within the QRS frequency band and attenuation of out-of-band content. Empirical end-to-end gain was determined from input/output voltage measurements with the ECG simulator (**Table 1**); a sample QRS amplitude of measured 6.5 mV peak-to-peak at the input produced 3.83 V peak-to-peak at the output, corresponding to a measured gain of 589.3 V/V. The comparator output for the frog electrogram along with the underlying electrogram signal is shown in **Figure 8**; each digital pulse aligns with a ventricular depolarization peak.

Acquisition and Processing Accuracy

R–R determination accuracy was assessed by feeding the ECG simulator at programmed rates of 60, 90, 120, and 220 BPM, with and without 10% R–R variability, and computing the device's measured BPM over a 10-second window. Results are summarized in **Table 2**. **Figure 9** shows pacemaker behavior with a 60-BPM ECG input and pacing programmed to 62 BPM (LRI = 0.968), barely above the natural pace. A paced pulse appears whenever no intrinsic beat occurs within the LRI. **Figure 10** shows the amplified ECG, comparator output, and pacing pulses at the 180-BPM upper-rate setpoint, where pacing intervals are tightly spaced near 0.333 s and sometimes occur immediately before intrinsic beats.

Stimulation

The strength–duration relationship was characterized in Module III by sweeping pulse duration and recording the threshold stimulus current. The data fit the expected hyperbolic form I = b(1 + c/d), yielding rheobase b = 1.5 µA and chronaxie c = 0.2 ms (see **Figure 11**). Based on these values,

the pacemaker stimulation block was programmed to deliver pulses of duration t_{pulse} = 2c = 0.4 ms at greater than twice the rheobasic current, providing a safety margin over the threshold curve. The output of the assembled MOSFET-driven stimulation circuit, recorded across the resistive load RL = 82.2 kΩ with a 10 µF storage capacitor and 82 kΩ charging resistor, is shown in **Figure 12**.

Added Functionality: Temperature-Responsive Rate Adaptation

Temperature was varied using a water-based cooling protocol: the thermistor and a separate reference thermometer were immersed in a beaker of hot water (initially approximately 80 °C, drawn from the laboratory water dispenser), and the water was progressively cooled by incremental addition of cold tap water. At each target temperature the system was allowed to equilibrate, after which the reference thermometer reading, the thermistor resistance (recorded with a digital multimeter), and the divider output V_{out} were captured. The voltage-divider relationship:

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{R_{\text{ref}}}{R_{\text{ref}} + R_{\text{therm}}}$$

was applied with V_{in} = 5 V and R_{ref} = 10 kΩ. Measured resistance across the thermistor and V_{out} increased monotonically with temperature, from 2.76 V at 30 °C to 3.55 V at 46 °C, with V_{out}(38 °C) ≈ 3.16 V at the febrile threshold (see **Figure 13** or **Appendix B.1** for the specific values and resistance values across the thermistor).

Pacing rate was then measured at each target water temperature, with the displayed BPM read directly off the OLED once the system equilibrated. Results are summarized in **Appendix B.2** and plotted in **Figure 14**. At water temperatures of 30 to 36 °C the device paced at the initial pacing baseline of 60 BPM. From 38 °C onward the pacing rate rose monotonically and saturated at the 180 BPM ceiling for water temperatures past 42 °C.

Graphical Data Presentation

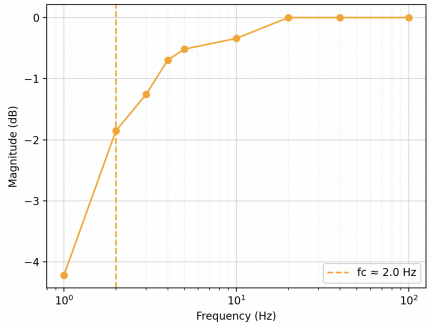


Figure 5. High-pass filter transfer function illustrating attenuation of low-frequency components (fc ≈ 2 Hz).

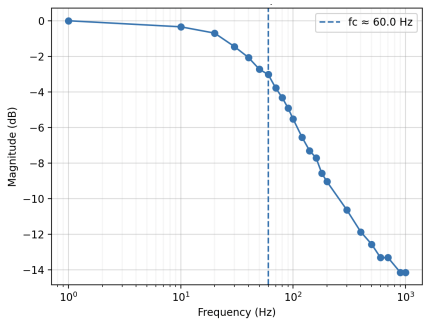


Figure 6. Low-pass filter transfer function illustrating attenuation of high-frequency noise (fc ≈ 60 Hz).

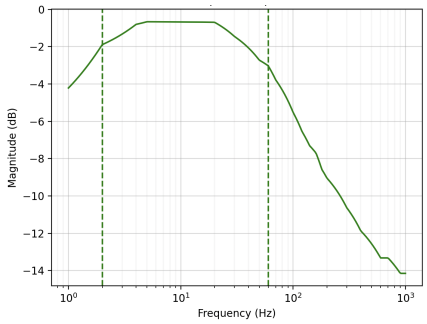


Figure 7. Cascaded bandpass response of the sensing circuit showing selective amplification of cardiac signal frequencies (~2–60 Hz passband).

Table 1. Empirical measurement of input voltage, output voltage, and calculated gain for the electrogram sensing circuit.

Vin (V)	Vout (V)	Gain
0.0065	3.83	589.3

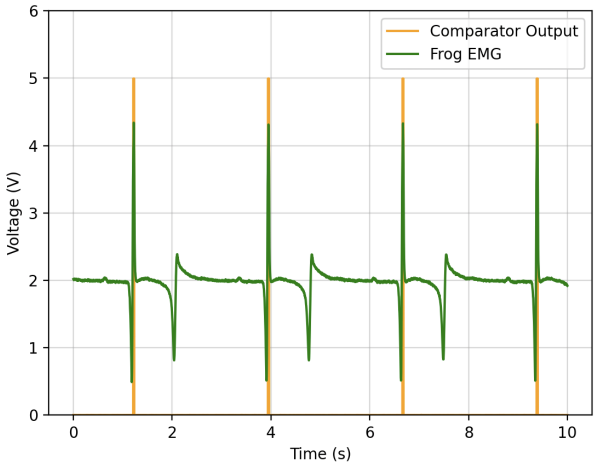


Figure 8. Comparator output overlaid on the conditioned electrogram, showing digital pulses generated at each detected ventricular depolarization.

Table 2. R–R determination accuracy across programmed heart rates with and without 10% R–R variability (10-second collection windows).

Defined Variability	**No R-R deviation**			**10% R-R deviation**	
Defined HR (BPM)	**60**	**90**	**120**	**60**	**220**
Mean Calculated HR (BPM)	60.12	90.28	124.62	56.39	220.7
Std. Dev. (BPM)	0.31	0.57	3.87	4.69	20.24
Percent Error (%)	0.20	0.31	3.85	6.01	0.34

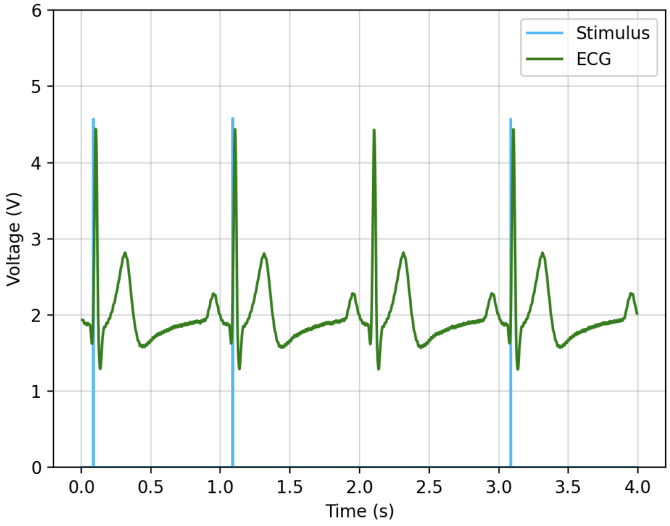


Figure 9. Pacemaker response with ECG at 62 BPM and pacing set to 60 BPM (LRI = 1 s), showing pacing when no natural beat occurs in time.

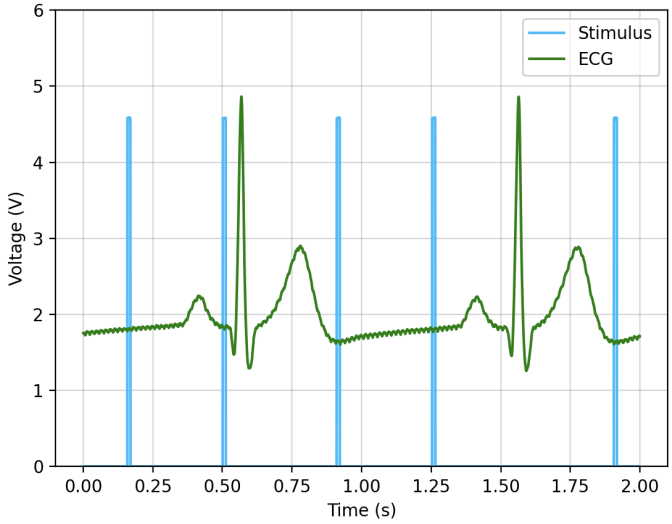


Figure 10. Amplified ECG and pacemaker pulses showing frequent pacing at ~0.333 s intervals (180 BPM setting), with pacing occurring close to intrinsic beats.

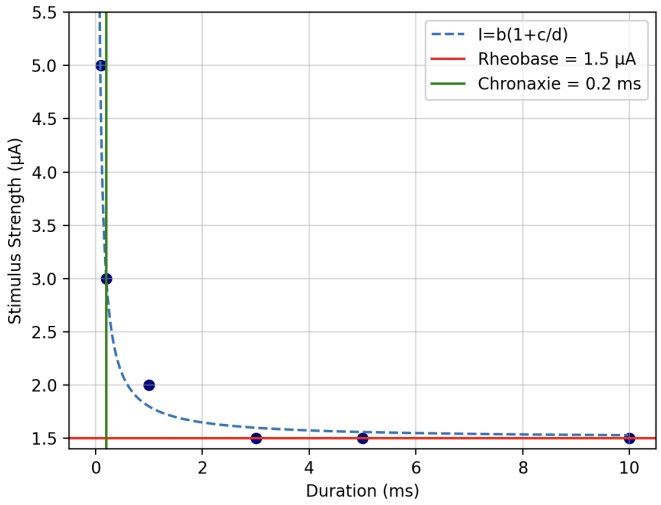


Figure 11. Strength–duration curve from Module III. Measured threshold currents (filled circles) fit I = b(1 + c/d) with rheobase b = 1.5 µA and chronaxie c = 0.2 ms. The stimulation operating point at (d_stim, I_stim) = (2c, 2b) = (0.4 ms, 3.0 µA) lies above the threshold curve, providing a safety factor for reliable capture.

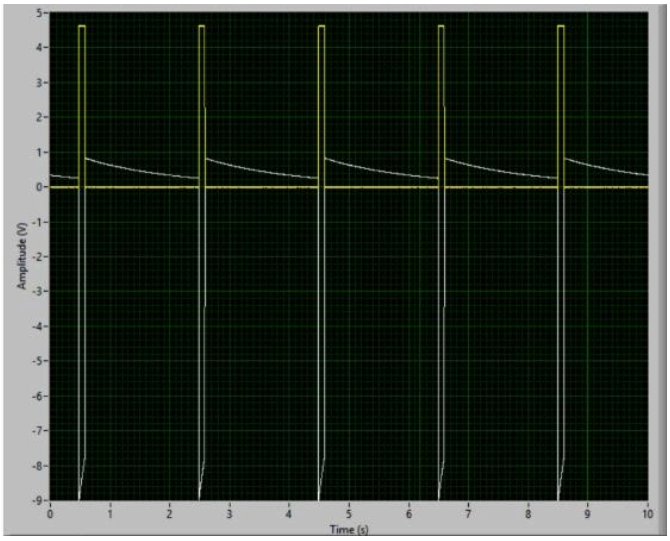


Figure 12. Stimulation circuit output across RL = 82.2 kΩ. Yellow trace: Arduino digital trigger to MOSFET gate. White trace: resulting stimulus pulse from capacitor discharge (C = 10 µF, Rc = 82 kΩ).

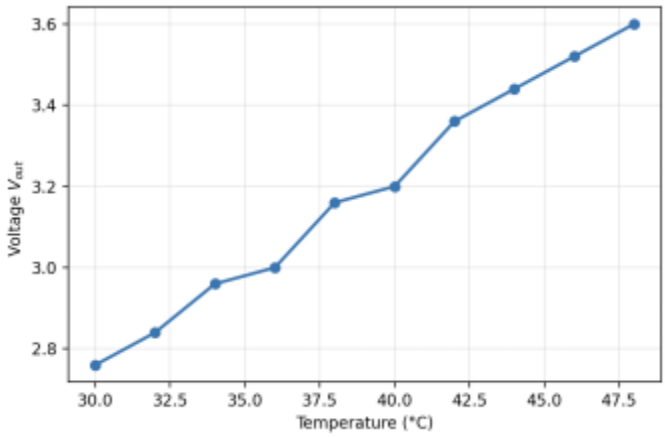


Figure 13. Thermistor voltage-divider output vs. temperature across 30–46 °C. V_out at the 38 °C threshold is 3.16 V.

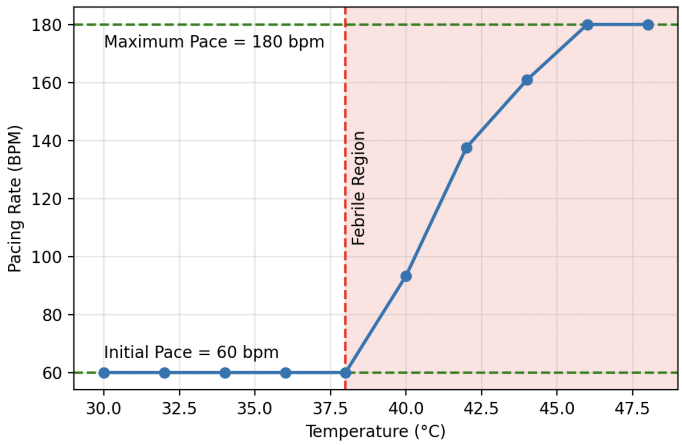


Figure 14. Measured pacing rate vs. bath temperature. The shaded region indicates the programmed febrile threshold (T > 38 °C); pacing baseline is 60 BPM and saturates at the 180 BPM ceiling.

DISCUSSION

The objective of this work was to design, build, and validate a temperature-responsive VVI-R pacemaker capable of sensing ventricular activity, pacing when intrinsic rhythm fell below a programmed lower rate, and adapting that rate to elevated body temperature. The integrated device performed each of these functions in benchtop testing and successfully paced the frog heart preparation under live electrogram input.

Design Performance

The sensing chain achieved a measured end-to-end gain of 589.3, sufficient to drive the millivolt-scale electrogram into a digital range where threshold detection at 3.72 V reliably tracked QRS events without spurious triggering on T-waves or baseline noise. R–R determination was accurate to within 4% across the physiologically relevant 60–120 BPM range with no induced variability, and the device continued to track rate at 220 BPM and under ±10% R–R variability — and there is a proportional standard deviation increase at the higher rates and variabilities. LRI-triggered pacing was demonstrated at programmed rates of 62 and 180 BPM, with paced pulses correctly inserted whenever intrinsic activity was absent within the LRI window. The MOSFET-driven capacitor-discharge stimulation block delivered pulses parameterized by the strength–duration data from our characterization (rheobase b = 1.5 µA, chronaxie c = 0.2 ms), with t_pulse = 2c and a 2× safety factor on current, the standard threshold-margin prescription for reliable myocardial capture.

Added Functionality

The rate-responsive temperature block extends the device beyond fixed-rate pacing into the VVI-R class. The 12.5 BPM/°C slope above 38 °C was selected to reproduce the well-documented physiological tachycardia of fever and is consistent with the rate–temperature relationships described in the rate-responsive pacing literature [2], [4], [5]. The 180 BPM ceiling is a deliberate safety bound: at sufficiently high temperatures the linear adaptation rule would otherwise prescribe heart rates above the safe physiological range, so the software enforces a hard cap to ensure that even with a runaway thermistor reading or a calibration fault the device cannot drive the heart into an unsafely fast rhythm. Clinically, fixed-rate pacing in a febrile patient under-paces relative to metabolic demand; the temperature sensor provides a low-cost, low-power proxy for that demand and closes the loop in a manner that motion- or ventilation-based sensors cannot for bedridden or critically ill patients. The OLED display, while not itself a clinical feature, served the practical role of giving the operator immediate visual confirmation that the closed loop was functioning during testing.

Summary and Future Work

In summary, a four-block VVI-R pacemaker was built from an instrumentation-amplifier sensing front end, an Arduino-based acquisition and LRI-processing subsystem, a MOSFET capacitor-discharge stimulation output, and a thermistor-driven rate-adaptation block. The device was characterized at every block, validated against an ECG simulator, and demonstrated to pace under live frog electrogram input. The MOSFET stimulation was undercharacterized, as only the empirical value for current across the frog was derived. Additional stimulus testing and changing the resistance in series to lower the current closer to twice the rheobase value would be a strong next step. One way we could increase the effectiveness of the temperature-adaptive pacing rate would be to average temperature over many time samples to avoid constantly changing the lower rate interval.

Additional future work would include replacing the discrete thermistor with a clinically appropriate core-temperature sensor; implementing hysteresis around the 38 °C threshold to prevent BPM oscillation near the febrile boundary; characterizing the device under physiologic inputs other than fever (e.g., exercise) by combining temperature with a complementary sensor such as an accelerometer; and extending characterization to longer (≥60 s) ECG-simulator runs to better resolve the variance behavior observed at high heart rates.

REFERENCES

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[2] E. Alt, "Temperature driven rate responsive cardiac pacemaker," U.S. Patent 4 688 573, Aug. 25, 1987.

[3] C. P. Lau et al., "Rate-adaptive cardiac pacing: Sensors, algorithms, and clinical applications," Archives of Cardiovascular Diseases, 2018.

[4] N. Sulke and A. J. Camm, "Sensors for rate responsive pacing," Pacing and Clinical Electrophysiology (PACE), vol. 16, 1993.

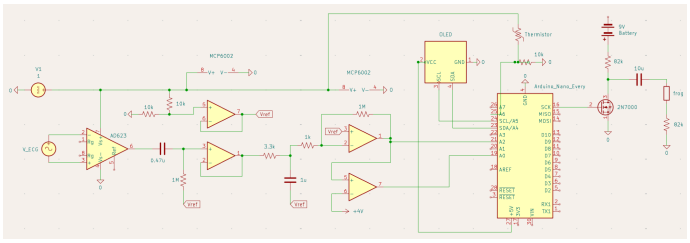
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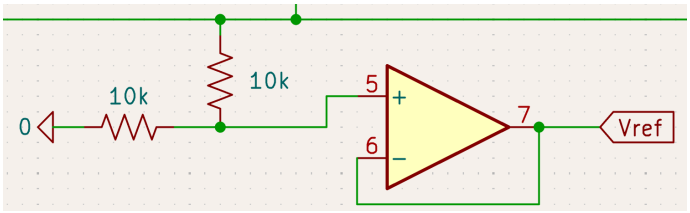
[6] Vishay BCcomponents, *NTC Thermistors, Radial Leaded, Standard Precision (NTCLE100E3)*. Malvern, PA, USA: Vishay Intertechnology, Inc, 2019.

APPENDIX

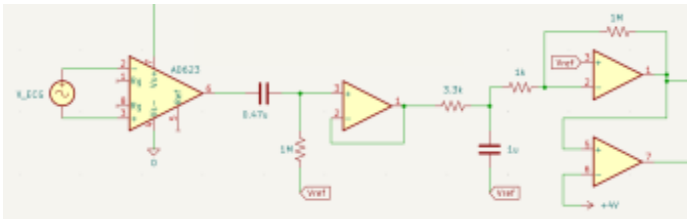
A Schematics



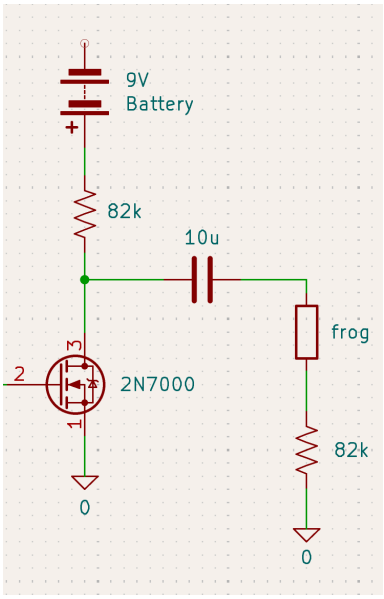
1- Full Schematic



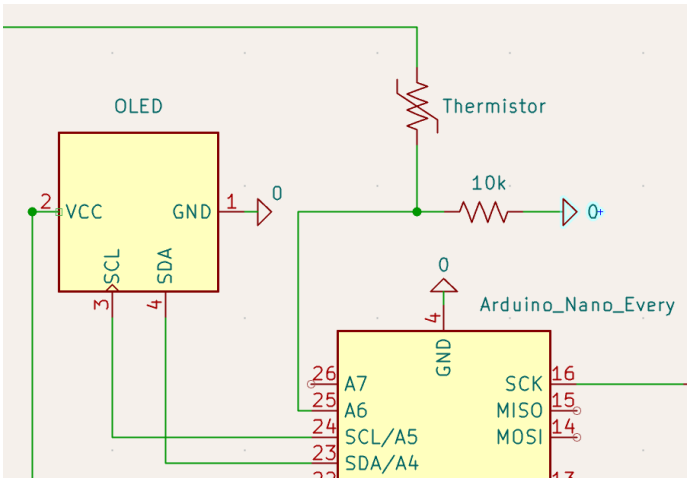
2- Zoomed Reference Voltage Schematic



3- Zoomed Sensing Schematic



4- Zoomed Stimulation Schematic



5- Zoomed Temperature and OLED Display Schematic

B Testing Results

1. Thermistor characterization: measured temperature, thermistor resistance (DMM), and voltage-divider output across 30–46 °C.

Temp (°C)	R_therm (kΩ)	V_out (V)
30	8.1	2.76
32	7.5	2.84
34	7.0	2.96
36	6.6	3.00
38	5.8	3.16
40	5.6	3.20
42	5.3	3.36
44	4.5	3.52
46	4.1	3.55
48	3.8	3.60

2. Measured pacing rate as a function of bath temperature, demonstrating the temperature-responsive rate-adaptation behavior with saturation at the 180 BPM ceiling.

Temp (°C)	Pacing HR (BPM)
30	60
32	60
34	60
36	60
38	60
40	93.3
42	137.6
44	160
46	180
48	180

C Microprocessor Code

Fully commented Arduino source code implementing R–R determination, LRI-based pacing logic, the temperature to BPM rate-adaptation rule, and OLED reporting is included below.

```
#include <Wire.h>

#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET     -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

const int ECG_amp_pin = 2;
const int ECG_comp_pin = 0;
const int I_stim_pin = 13;
const int temp_pin = 20; // voltage from thermistor circuit
float ECG_ampV = 0.0, ECG_compV = 0.0, temp_V = 0.0;
unsigned long prev_RR = 0, current_RR = 0;
float HR = 0.0, display_bpm;
int ts = 1; // sampling period (ms)
unsigned long prevtime = 0;
int display_s = 2000;
unsigned long prev_display = 0;

float i_bpm = 62.0;
int i_LRI = 60000 / i_bpm;
int LRI = i_LRI;
float i_temp = 36.0; // C
float delta_T = 0.0, delta_bpm = 0.0, bpm = 0.0;
float chronaxie = 0.200; // ms (from Module III)

// Thermistor / Steinhart–Hart (4-coefficient form)
float R1 = 10000.0, Vin = 5.0, Rt = 0.0;
const float t_A1 = 0.003354016, t_B1 = 0.000256985;
const float t_C1 = 0.000002620131, t_D1 = 0.00000006383091;
const float Rref = 10000.0;
float lnTerm, T_kelvin, T;

bool beatDetected = false;

void setup() {
 Serial.begin(115200);
 pinMode(I_stim_pin, OUTPUT);
 if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) { for (;;); }
 display.clearDisplay(); display.display();
}

void loop() {
 if (millis() - prevtime >= ts) {
      int ECG_comp = analogRead(ECG_comp_pin);
      ECG_compV = ECG_comp * 5.0 / 1023;

      // QRS detection on comparator output
      if (!beatDetected && ECG_compV > 2.5) {
      beatDetected = true;
      current_RR = millis();
      HR = 60000.0 / (current_RR - prev_RR);
      display_bpm = HR;
      prev_RR = current_RR;
      } else if (ECG_compV < 1.0) {
      beatDetected = false;
      }

      // LRI-triggered pacing
      if (millis() - prev_RR > LRI) {
      digitalWrite(I_stim_pin, HIGH);
      delay(2 * chronaxie);
      digitalWrite(I_stim_pin, LOW);
      }

      // Thermistor → Steinhart–Hart → temperature
      int temp_input = analogRead(temp_pin);
      temp_V = temp_input * 5.0 / 1023;
      Rt = R1 * (Vin - temp_V) / temp_V;
      lnTerm = log(Rt / Rref);
      T_kelvin = 1.0 / (t_A1 + t_B1 * lnTerm + t_C1 * lnTerm * lnTerm
                        + t_D1 * lnTerm * lnTerm * lnTerm);
      T = T_kelvin - 273.15;

      // Temperature-responsive rate adaptation
      delta_T = T - i_temp;
      if (delta_T > 2.0) {                // T > 38 C: fever state
      delta_bpm = 12.5 * delta_T;
      bpm = i_bpm + delta_bpm;
      if (bpm > 180.0) bpm = 180.0;                // cap
      LRI = 60.0 / bpm * 1000;
      } else {                            // normal
      bpm = i_bpm;
      LRI = i_LRI;
      }
      prevtime = millis();
 }

 // OLED update at display_s interval
 if (millis() - prev_display > display_s) {
      display.clearDisplay();
      display.setTextColor(SSD1306_WHITE);
      display.setTextSize(1); display.setCursor(0, 0); display.print("Temp: ");
      display.setTextSize(2); display.print(T, 1);
      display.setTextSize(1); display.print(" C");
      display.setCursor(0, 22); display.print("Target HR: ");
      display.setTextSize(2); display.print(bpm, 1);
      display.setTextSize(1); display.setCursor(0, 32); display.print("(bpm)");
      if (delta_T > 2.0) { display.setTextSize(2); display.setCursor(0, 45);
display.print("Fever"); }
      else { display.setTextSize(1); display.setCursor(0, 50); display.print("Normal"); }
      display.display();
      prev_display = millis();
 }
}
```